Topic : Naive Bayes

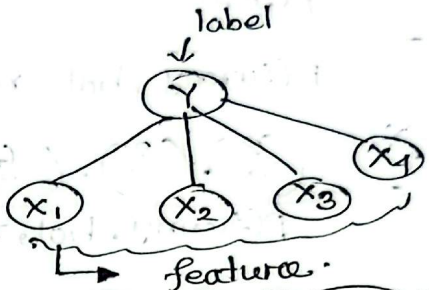
i) $P(x/y) = \frac{P(x,y)}{P(y)}$ → Conditional probability (already known)

ii) $P(x,y) = P(x/y)\,P(y) = P(y/x)\,P(x)$ → joint probability or product rule

$\Rightarrow P(x/y) = \frac{P(y/x)\,P(x)}{P(y)}$ * → naive Baye's rule

x_1	x_2	x_3	x_4	Y
height	weight	tail	shape	cat/dog
⋮	⋮	⋮	⋮	⋮

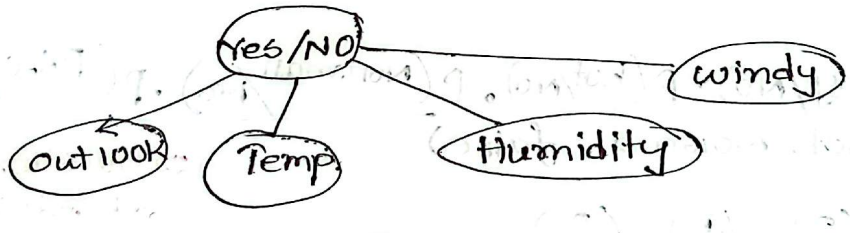
features ← | label



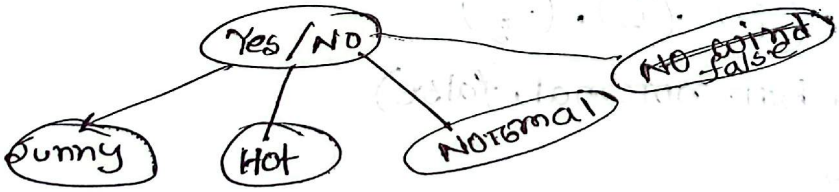
* যতগুলো label থাকবে সবগুলোর জন্য probability বের করবো। যার probability বেশি তাকে accept করবো।

feature গুলো label এর উপর dependent.

Example : Today is sunny, Hot, Humidity = Normal, and NO wind. what is probability of playing golf?



যার parent নেই সে direct বসবে।



$P(Yes / sunny, Hot, Normal, False)$

$= \frac{P(yes, sunny, hot, normal, false)}{P(sunny, hot, normal, false)}$ → product rule

$\Rightarrow \frac{P(sunny/yes) \cdot P(hot/yes) \cdot P(normal/yes) \cdot P(false/yes) \cdot P(yes)}{P(sunny, hot, normal, false)}$

(yes এর মধ্যে কথাটা sunny? | yes এর মধ্যে কথাটা hot? | yes এর মধ্যে normal কথাটা? | এখানে ব্যবহৃত হবে কারণ parent টেবিল।)

$= \frac{\left(\frac{3}{9}\right) \cdot \left(\frac{2}{9}\right) \cdot \left(\frac{6}{9}\right) \cdot \left(\frac{6}{9}\right) \cdot \left(\frac{9}{14}\right)}{P(sunny, hot, normal, false)}$

$= \frac{0.021}{P(sunny, hot, normal, false)}$

decision কোথায় থেকেই নিতে পারবো যে খেলবে কি খেলবে না, আর যদি exact probability চায় তাহলে বের করা লাগবে।

again,

$P(NO / sunny, hot, Normal, false)$

$= \frac{P(NO, sunny, hot, Normal, false)}{P(sunny, hot, Normal, false)}$

$= \frac{P(NO) \cdot P(sunny/NO) \cdot P(hot/NO) \cdot P(Normal/NO) \cdot P(False/NO)}{P(sunny, hot, Normal, false)}$

এই Question এ probability বের করা লাগবে।

$= \frac{\left(\frac{5}{14}\right) \cdot \left(\frac{2}{5}\right) \cdot \left(\frac{2}{5}\right) \cdot \left(\frac{1}{5}\right) \cdot \left(\frac{2}{5}\right)}{P(sunny, hot, normal, false)}$

$= \frac{0.005}{P(sunny, hot, normal, false)}$

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$\because P(\text{Sunny, hot, Normal, false}) = P(\text{yes, sunny; hot, Normal, false}) +$
$P(\text{No, sunny, hot, Normal, false})$

$= 0.021 + 0.005$

$= 0.026$

$\because P(\text{yes/sunny, hot, Normal, false}) = \frac{0.021}{0.026} = 0.8079$

$P(\text{No/sunny, hot, Normal, false}) = \frac{0.005}{0.026} = 0.1923$

Example:- The weather data example.



Play?

$$P(\text{Yes}/S, C, H, T) = \frac{P(\text{yes}, S, C, H, T)}{P(S, C, H, T)}$$

$$= \frac{P(\text{yes}) \cdot P(S/\text{yes}) \cdot P(C/\text{yes}) \cdot P(H/\text{yes}) \cdot P(T/\text{yes})}{P(S, C, H, T)}$$

$$= \frac{\frac{9}{14} \cdot \frac{2}{9} \cdot \frac{3}{9} \cdot \frac{3}{9} \cdot \frac{3}{9}}{P(S, C, H, T)}$$

$$= \frac{0.0053}{P(S, C, H, T)}$$

$$P(NO/S,C,H,T) = \frac{P(NO,S,C,H,T)}{P(S,C,H,T)}$$

$$= \frac{P(NO)\cdot P(S/NO)\cdot P(C/NO)\cdot P(H/NO)\cdot P(T/NO)}{P(S,C,H,T)}$$

$$= \frac{\frac{5}{11}\cdot\frac{3}{5}\cdot\frac{1}{5}\cdot\frac{1}{5}\cdot\frac{3}{5}}{P(S,C,H,T)} = \frac{0.0206}{P(S,C,H,T)}$$

More probable "NO"

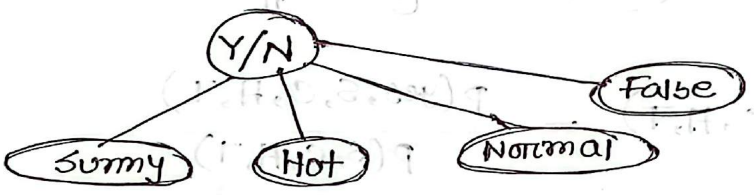
"Smoothing"

Label এর against এ feature এর value zero হলে probability zero হয়ে যায় যা ভুল। তাই এড়াতে smoothing করা হয়।

- ভুলে যাবেই চিন্তা করবেনা feature এর value zero হবে না।
- smoothing factor(k) question এ বলা থাকে। [K=1,2.....]

Example:

K=2



$$P(Y/S,H,N,F) = \frac{P(Y,S,H,N,F)}{P(S,H,N,F)}$$

$$= \frac{P(Y)\cdot P(S/Y)\cdot P(H/Y)\cdot P(N/Y)\cdot P(F/Y)}{P(S,H,N,F)}$$

$$= \frac{\frac{11}{14+(2*2)}\cdot\frac{5}{9+2*3}\cdot\frac{4}{9+2*3}\cdot\frac{8}{9+2*2}\cdot\frac{8}{9+2*2}}{P(S,H,N,F)}$$

(K=2 → 11; K=2 → 5; K=2 → 4; K=2 → 8; K=2 → 8)

- Play Golf এর মধ্যে 2 টা option, তাই প্রত্যেকের জন্য K=2 করে বাড়বে। ∴ option × K
- no. of option এর সাথে K গুন।
- Y এর জন্য 2 টা, N এর জন্য 2 টা
- outlook এর option 3 টা → K
- Tem এর option 3 টা → K
- Humidity এর option 2 টা → K
- Windy এর option 2 টা → K

$$= \frac{0.021}{P(S,H,N,F)}$$

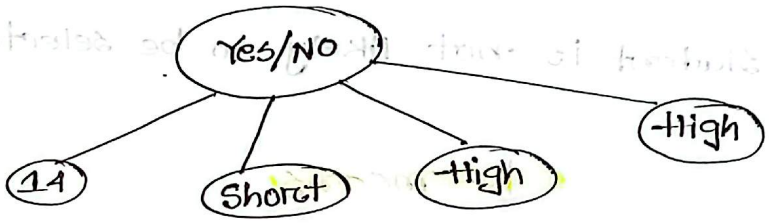
$$P(N/S,H,N,F) = \frac{P(N)\cdot P(S/N)\cdot P(H/N)\cdot P(N/N)\cdot P(F/N)}{P(S,H,N,F)}$$

$$= \frac{\frac{5+2\to K}{14+2*2}\cdot\frac{2+2\to K}{5+2*3}\cdot\frac{2+2\to K}{5+2*2}\cdot\frac{1+2\to K}{5+2*2}\cdot\frac{2+2\to K}{5+2*2}}{P(S,H,N,F)}$$

$$= \frac{0.0093}{P(S,H,N,F)}$$

∴

Practice (Slide – 33)



$K = 1.$

$$P(Yes/14,S,H,H) = \frac{P(yes,14,S,H,H)}{P(14,S,H,H)}$$

$$= \frac{P(yes)\times P(14/yes)\times P(S/yes)\times P(H/yes)\times P(H/yes)}{P(14,S,H,H)}$$

$$= \frac{\frac{5+1}{10+1\times2}\times\frac{2+1}{5+1\times3}\times\frac{1}{5+1*3}\times\frac{3+1}{5+1*2}\times\frac{4+1}{5+1*2}}{P(14,S,H,H)}$$

$$= \frac{\frac{6}{12}\times\frac{3}{8}\times\frac{1}{8}\times\frac{4}{7}\times\frac{5}{7}}{P(14,S,H,H)} = \frac{0.0096}{P(14,S,H,H)}$$

$$P(N/14,S,H,H) = \frac{P(N,14,S,H,H)}{P(14,S,H,H)}$$

$$= \frac{P(N) \times P(14/N) \times P(S/N) \times P(H/N) \times P(H/N)}{P(14,S,H,H)}$$

$$= \frac{\frac{5+1}{10+1*2} * \frac{2+1}{5+1*3} \times \frac{3+1}{5+1*3} \times \frac{2+1}{5+1*2} \times \frac{1+1}{5+1*2}}{P(14,S,H,H)}$$

$$= \frac{\frac{6}{12} \times \frac{3}{8} \times \frac{4}{8} \times \frac{3}{7} \times \frac{2}{7}}{P(14,S,H,H)}$$

$$= \frac{0.011}{P(14,S,H,H)}$$

∴ The student is not likely to be selected. (Ans)

K-means

problem: Classify the following data into two clusters using the k-means algorithm (3,4), (6,7) (4,5) (5,7) (2,6)

সেই কয়টা clusters বলবে সেই কয়টা class নিবো centroid হিসেবে.

Soln: Iteration 1:

let class 1 : (3,4)
class 2 : (4,5)

Data	class-1 (3,4)	class-2(4,5)	class
(3,4)	0	$\sqrt{(3-4)^2+(4-5)^2} = 1.414$	class-1 ✓
(6,7)	4.242	2.828	class-2
(4,5)	1.414	0	class-2
(5,7)	3.61	2.236	class-2
(2,6)	2.236	2.236	class-1 ✓

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∴ class 1 : (3,4) (2,6) $= \left(\frac{3+2}{2}, \frac{4+6}{2}\right) = \left(\frac{5}{2}, \frac{10}{2}\right) = (2.5, 5)$

class 2 : (6,7) (4,5) (5,7) $= \left(\frac{6+4+5}{3}, \frac{7+5+7}{3}\right) = \left(\frac{15}{3}, \frac{19}{3}\right) = (5, 6.33)$

Iteration 2: New centroid,

(*) $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

class-1 : (2.5, 5)

class-2 : (5, 6.33)

Data	class-1 (2.5,5)	Class-2 (5, 6.33)	Class
(3,4)	1.12	3.07	class-1
(6,7)	1.03	1.204	Class-2
(4,5)	1.5	1.66	class-1
(5,7)	3.20	0.7	Class-2
(2,6)	1.12	3.01	Class-1

∴ class-1 : (3,4) (4,5) (2,6) $= \left(\frac{3+4+2}{3}, \frac{4+5+6}{3}\right) = (3, 5)$

class-2 : (6,7) (5,7) $= \left(\frac{6+5}{2}, \frac{7+7}{2}\right) = (5.5, 7)$

Iteration-3: New centroid,

class-1 : (3,5)

Class-2 : (5.5,7)

Data	class-1 (3,5)	class-2 (5.5,7)	class
(3,4)	1	3.90	class-1
(6,7)	3.61	0.5	class-2
(4,5)	1.0	2.5	class-1
(5,7)	2.83	0.5	class-2
(2,6)	1.414	3.64	class-1

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Class-1 : $(3,4)\ (4,5)\ (2,6) = \left(\frac{3+4+2}{3}, \frac{4+5+6}{3}\right)$

$= (3,5)$

Class-2 : $(6,7)\ (5,7) = \left(\frac{6+5}{2}, \frac{7+7}{2}\right)$

$= (5.5, 7)$

previous centroid = current centroid so algorithm stop.

output:

Data	class	centroid
(3,4) (4,5) (2,6)	class - I	(3,5)
(6,7) (5,7)	class - II	(5.5, 7) .

Steps: → cluster যেই কয়টা বলবে সেই কয়টা class এ ভাগ করতে হবে এবং initially সেই কয়টা centroid নিবে।

→ centroid গুলো থেকে প্রত্যেকটা pair এর euclidean distance $\left(\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}\right)$ বের করবো। centroid থেকে যার distance কম থাকে সেই class এ দিবো।

→ তারপর সেই class এ যেই কয়টা pair হবে তাদের average বের করবো। এই average value-ই হবে নতুন centroids.

→ ~~previous~~ updated centroids যদি ~~updated~~ previous centroids এর মতো হয় তাহলে algorithm stop করে দিবো।

→ এখন যদি কোনো একটা data দিয়ে বলে এটা কোন class এ পড়বে তাহলে, updated centroids থেকে ঐ given data এর distance বের করবো। যেই class এর জন্য distance কম আসবে তাকে ঐ class এ দিবো।

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For example, classify this data (2,7) using the previous experiment.

Soln: updated centroids

class-1 : (3,5) । এখন এইটা এবং (2,7) এর distance বের করবো।

class-2 : (5.5,7)। " " এবং " " " " "

যেই centroid এর সাথে distance কম হবে, (2,7) কে ঐ class এ দিবো।

class-1: (3,5) ; $d_1 = \sqrt{(3-2)^2+(5-7)^2} = 2.24$

class-2 : (5.5,7); $d_2 = \sqrt{(5.5-2)^2+(7-7)^2} = 3.5$

Here 2.24 is smaller than 3.5 . so the given Data (2,7) is in class-1 . (Ans)

Modified K-means.

↳ centroid randomly ধরবো না।
সব pair এর সাথে সব pair এর distance বের করবো। then large distance select করবো। largest distance যেসব pair এর জন্য হয়েছে তাদেরকে centroids ধরবো।

Question: Classify the following data into two clusters using the modified k-means algorithm: (3,4) (6,7) (4,5) (5,7) (2,6)

Soln: initial centroid selection,

Data	(3,4)	(6,7)	(4,5)	(5,7)	(2,6)
(3,4)	0	(4.243)	1.41	3.61	2.24
(6,7)	4.243	0	2.83	1	4.123
(4,5)	1.41	2.83	0	2.236	2.236
(5,7)	3.61	1	2.236	0	3.162
(2,6)	2.24	4.123	2.236	3.162	0

Class-1 : (3,4)

Class-2 : (6,7)

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Iteration 1:

Data	class-1 (3,4)	class-2 (6,7)	class
(3,4)	0	4.242	class-1
(6,7)	4.242	0	class-2
(4,5)	1.414	2.828	class-1
(5,7)	3.61	1	class-2
(2,6)	2.236	4.123	class-1

∴ class-1 : (3,4) (4,5) (2,6) $= \left(\frac{3+4+2}{3}, \frac{4+5+6}{3}\right) = (3,5)$

class-2 : (6,7) (5,7) $= \left(\frac{6+5}{2}, \frac{7+7}{2}\right) = (5.5, 7)$

Iteration 2: New centroids,

class-1 : (3,5)

class-2 : (5.5, 7)

Data	class-1 (3,5)	class-2 (5.5,7)	class
(3,4)	1	3.91	class-1
(6,7)	3.61	0.5	class-2
(4,5)	1	2.5	class-1
(5,7)	2.83	0.5	class-2
(2,6)	1.414	3.64	class-1

∴ class-1 : (3,4) (4,5) (2,6) $= \left(\frac{3+4+2}{3}, \frac{4+5+6}{3}\right) = (3,5)$

class-2 : (6,7) (5,7) $= \left(\frac{6+5}{2}, \frac{7+7}{2}\right) = (5.5, 7)$

∴ algorithm stop

output:

Data	class	centroid
(3,4)(4,5)(2,6)	class-1	(3,5)
(6,7)(5,7)	class-2	(5.5, 7)

(Ans)

" Monte Carlo "

During the past (200) days the demand for the car fluctuated as below

Trips per week	Frequency
0	16
1	24
2	30
3	60
4	40
5	30

Simulate the demand for a 10-week period use the random no:
82, 96, 18, 96, 20, 84, 56, 11, 52, 3

Soln :- Step 1 : probability, cumulative probability, range বের করবো।

N.B : Question এ probability দেয়া থাকলে Cumulative probability and range বের করবো।

Step 2 : যেই কয়দিনের জন্য random numbers দেয়া থাকবে সেই random numbers কোন range এ পরে তার corresponding demand মানে Trips per week বের করবো।

Step 3 : তারপর Average demand বের করবো।

Trips per week/ Demand	Frequency	probability	cumulative probability (100 দিয়ে multiply করবো।)	Tag-numbers/Range.
0	16	16/200 = 0.08	0.08	0 - 7
1	24	24/200 = 0.12	0.2	8 - 19
2	30	30/200 = 0.15	0.35	20 - 34
3	60	60/200 = 0.3	0.65	35 - 64
4	40	40/200 = 0.2	0.85	65 - 84
5	30	30/200 = 0.15	1	85 - 99

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Now,

week	Random numbers	Demand
1	82	4
2	96	5
3	18	1
4	96	5
5	20	2
6	84	4
7	56	3
8	11	1
9	52	3
10	03	0

Total = 28

$\therefore$ Average Demand $= \frac{28}{10} = 2.8$ cars/week

Problem

*Q2.

Production/day	probability
95	0.03
96	0.05
97	0.07
98	0.10
99	0.15
100	0.20
101	0.15
102	0.10
103	0.07
104	0.05
105	0.03

Finished cars are transported across the bay at the end of each day by a ferry. If ferry has space for only 101 cars, what will be the average no. of cars waiting to be shipped and what will be the average no. of empty space on the ship. Use the random: 97, 02, 80, 66, 96, 55, 50, 29, 58, 51, 01, 86, 24, 34, 47.

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Soln:

production/day	probability	cumulative probability	Tag - numbers.
95	0.03	0.03	0-2
96	0.05	0.08	3-7
97	0.07	0.15	8-14
98	0.10	0.25	15-24
99	0.15	0.4	25-39
100	0.20	0.6	40-59
101	0.15	0.75	60-74
102	0.10	0.85	75-84
103	0.07	0.92	85-91
104	0.05	0.97	92-96
105	0.03	1	97-99

The simulated production of cars for the next 15 days is given in the following table:

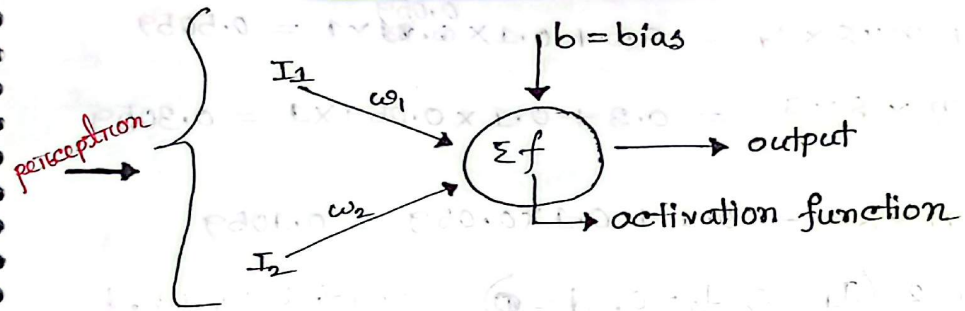
Maximum capacity 101

Days	Random numbers	Demand production per day	No. of cars waiting.	No. of empty space in ship.
1	97	105	1	—
2	02	95	—	6
3	80	102	1	—
4	66	101	—	—
5	96	104	3	—
6	55	100	—	1
7	50	100	—	1
8	29	99	—	2
9	58	100	—	1
10	51	100	—	1
11	04	96	—	5
12	86	103	2	—
13	24	98	—	3
14	34	99	—	2
15	47.	100	—	1
			Total = 10	total = 23

∴ Average number of cars waiting to be shipped = $\frac{10}{15}$ = 0.67 per day.

∴ Average no. of empty space on the ship = $\frac{23}{15}$ = 1.53 per day.

"Artificial Neural Network (ANN)"



ω = weight (কোন input এর priority বেশি তা decide করে)

learning rate (η)

d = expected output.

Sigmoid function $= \frac{1}{1+e^{-v}}$ ← weighted sum

problem

Training data

Input		output
I_1	I_2	d
1	1	1
0	0	0
0	1	??

Initial weight, $\omega_1 = 0.5$
$\omega_2 = 0.3$
$b = 0.1$

Test data 0 & and 1
Find output for test data, where learning rate (η) = 0.1.

Sol^n: For training data:

Forword pass: pattern 1 ($I_1 = 1, I_2 = 1, d = 1$)

∴ weighted sum, $v = I_1\omega_1 + I_2\omega_2 + b$
$= 1 \times 0.5 + 1 \times 0.3 + 0.1$
$= 0.9$

$$f = \frac{1}{1+e^{-v}} = \frac{1}{1+e^{-0.9}} = 0.71$$ ← calculated output

actual output 1 কিন্তু calculated output পেলাম 0.71, তাই এখানে adjust করা লাগবে। এই adjustment এর জন্যই back propagation করা লাগে।

Back propagation:

কতটুকু adjustment দরকার।

$\delta = f(1-f)(d-f)$
$= 0.71(1-0.71)(1-0.71) = 0.059$

weight update

given

$w_1 = w_1 + \Delta w_1 = w_1 + \eta \times \delta \times I_1 = 0.5 + 0.1 \times 0.059 \times 1 = 0.5059$

$w_2 = w_2 + \Delta w_2 = w_2 + \eta \times \delta \times I_2 = 0.3 + 0.1 \times 0.059 \times 1 = 0.3059$

given

$b = b + \Delta b = b + \eta \times \delta = 0.1 + 0.1 \times 0.059 = 0.1059$

Forward pass: pattern 2 ($I_1 = 0$, $I_2 = 0$, $d = 0$) → updated w_1, w_2, b use করবো।

$v = w_1 I_1 + w_2 I_2 + b$

$= 0.5059 \times 0 + 0.3059 \times 0 + 0.1059$

$= 0.1059$

$f = \frac{1}{1+e^{-v}} = \frac{1}{1+e^{-0.1059}} = 0.526$ → calculated output

actual output আমাদের কথা zero কিন্তু আমরা 0.526 আর্থ adjust করা লাগবে

$\delta = f(1-f)(d-f)$

$= 0.526\,(1-0.526)\,(0-0.526)$

$= -0.1311$

weight update:

$w_1 = w_1 + \Delta w_1 = w_1 + \eta \times \delta \times I_1 = 0.5059 + 0.1 \times (-0.1311) \times 0$

$= 0.5059$

$w_2 = w_2 + \Delta w_2 = w_2 + \eta \times \delta \times I_2 = 0.3059 + 0.1 \times (-0.1311) \times 0$

$= 0.3059$

$b = b + \Delta b = b + \eta \times \delta = 0.1059 + (-0.1311) \times 0.1$

$= 0.09279$

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Testing : (Testing এ back propagation নেই) [updated w_1, w_2, b use করবো]

$I_1 = 0$
$I_2 = 1$
$d = ?$

$$v = w_1 I_1 + w_2 I_2 + b$$
$$= 0.5059 \times 0 + 0.3059 \times 1 + 0.09279$$
$$= 0.39869$$

$$\therefore f = \frac{1}{1+e^{-v}} = \frac{1}{1+e^{-0.39869}} = 0.5983 \text{ (Ans)}$$

"Regression"

→ কোনো কিছু predict করা।
→ linear regression (m,c. বের করা)
→ Logistic : probability হিসেবে output পাবো
regression

Linear regression

Example :

X week	Y sales
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

যার মানে average

$$c = \bar{y} - m\bar{x}$$

Linear regression equation is given

$$y = a_0 + a_1 * x + e$$
$$\downarrow$$
$$y = mx + c$$

where,

$$m = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2}$$

Apply linear regression technique to predict the 7th and 12th week sales.

Soln :

Week x	Sales y	x^2	xy
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15	12.6	55	44.4
average $\bar{x} = 15/5 = 3$	$\bar{y} = \frac{12.6}{5} = 2.52$	$\overline{x^2} = \frac{55}{5} = 11$	$\overline{xy} = 44.4/5 = 8.88$

$$\therefore m = \frac{\overline{xy} - \bar{x}\cdot\bar{y}}{\overline{x^2} - \bar{x}^2}$$

$$= \frac{8.88 - 3 \times 2.52}{11 - (3)^2}$$

$$= 0.66$$

$$c = \bar{y} - m\bar{x}$$

$$= 2.52 - 0.66 \times 3$$

$$= 0.54$$

$\therefore$ Regression equation is, $y = 0.66x + 0.54$

when $x = 7$

$\therefore y = 0.66 \times 7 + 0.54$

$= 5.16$

when $x = 12$

$\therefore y = 0.66 \times 12 + 0.54$

$= 8.46$ (Ans)

Logistic Regression

↳ probability হিসেবে output চাইলে এইটা use করবো।

Sigmoid function equation : $S(z) = \frac{1}{1+e^{-z}}$; $z = mx + c$

↳ [0, 1] এর মধ্যে যেকোনো value হতে পারে।

(1)

Hours study	pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

z এর equation দেয়া থাকবে।

$z = -64 + 2 * hours$

calculate the probability of pass for the student who studied (33) hours.

Soln: given ; $z = -64 + 2 * hours$

$= -64 + 2 * 33$

$= 2$

$\therefore S(z) = \frac{1}{1+e^{-2}}$

Probability (P) $= 0.88$

Sigmoid function এর threshold value always 0.5। এর নিচে গেলে fail করবে আর উপরে গেলে pass।

∴ If the student studies 33 hours then there is 88% chance that the student will pass the exam.

$x = ?$

(2) At least (how) many (hours) student should study that makes he will pass the course with the probability of more than 95%

Soln: given, $p = 0.95$

$$\Rightarrow \frac{1}{1+e^{-z}} = 0.95$$

$$\Rightarrow (1+e^{-z}) \times 0.95 = 1$$

$$\Rightarrow 0.95 + 0.95 \times e^{-z} = 1$$

$$\Rightarrow e^{-z} = \frac{1-0.95}{0.95}$$

$$\Rightarrow \ln(e^{-z}) = \ln\left(\frac{1-0.95}{0.95}\right)$$

$$\Rightarrow -z = -2.944$$

$$\Rightarrow z = 2.944$$

$\because z = -64 + 2 * \text{hours}$

$\Rightarrow 2.944 = -64 + 2 * \text{hours}$

$\Rightarrow 2 * \text{hours} = 2.944 + 64$

$\Rightarrow \text{hours} = \frac{2.944+64}{2}$

$\Rightarrow \text{hours} = 33.472$ hours.

$\therefore$ The student should study at least 33.47 hours so that he will pass the exam with more than 95% probability.

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